

OpenEd Problem-Solution Document

Problem statement

Online learning has become abundant, but serious skill formation is still hard. Learners can access thousands of videos and courses, yet many still do not know what to learn, how to verify what they understood, how to get doubts solved, or how to prove ability. Educators can upload content, but the platforms rarely force good instructional structure, source mapping, or meaningful proof of learning.

OpenEd targets the gap between content access and actual learning.

Problem tree

Learner problems

Symptom	Root problem	Consequence
Video fatigue	Lessons are passive	Learners stop opening the course.
Doubts remain unresolved	AI tutor or human mentor is missing/paid	Learners lose momentum.
Too many resources	No learning path or source hierarchy	Learners jump between links.
Weak certificates	Completion is not proof	Employers/peers cannot trust claims.
Poor feedback	Quizzes are shallow or absent	Learners do not know what to improve.
Language/confidence gap	Content assumes fluent English and prior context	Learners feel excluded.

Educator problems

Symptom	Root problem	Consequence
Good educators cannot publish well	Course creation tools are too limited or too scattered	Knowledge stays in notes and videos.
Anyone can upload low-quality content	Platform checks focus on upload, not teaching proof	Learner trust drops.
	Rubrics and artifacts are not native	

Symptom	Root problem	Consequence
Assessments are hard to design		Courses become content dumps.
Sources are not mapped	No structure for citations and advanced resources	Learners cannot go deeper reliably.

Platform problems

Symptom	Root problem	Consequence
Low completion	Product optimizes catalog size over learning loop	Users churn.
AI feature is generic	Chatbot is bolted onto content	Answers are not trustworthy or pedagogical.
Grading is either too weak or too heavy	No flexible proof levels	Certificates lose meaning.
Mobile and desktop conflict	One layout is stretched across devices	Learning becomes uncomfortable.

Solution

OpenEd provides a Learn-Prove system.

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Choose goal -> diagnostic -> learning path -> lesson -> tutor -> practice -> artifact -> feedback -> SkillProof portfolio
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The product does not treat content as the course. A course is a structured learning environment made of:

- Lessons.
- Sources.
- Tutor prompts.
- Practice tasks.
- Quizzes.
- Rubrics.
- Artifacts.
- Feedback.
- Portfolio output.

Solution pillars

1. Source-mapped courses

Every serious course should show where its claims come from. Sources are not an appendix; they are part of the learner's trust model.

Source cards include title, author/organization, type, difficulty, required/optional status, why it matters, and recommended section or timestamp.

2. Course-grounded AI tutor

The AI tutor answers from the course context, lesson summary, transcript, source cards, artifact rubric, and learner state. It should not act like a generic chatbot that simply produces answers.

3. Practice and artifacts

Every meaningful module should include a proof output: a memo, design, notebook, rubric, project, model, report, or similar artifact. Artifacts become the learner's SkillProof portfolio.

4. Educator Studio

Educators create courses using structured building blocks: outline, lesson, source cards, media, quiz, practice, rubric, artifact, publish-readiness QA.

5. Assessment and Proof Engine

OpenEd separates completion from mastery. The proof ladder is:

Level	Meaning
Seen	Learner opened or watched/read content.
Checked	Learner passed a knowledge check.
Practiced	Learner completed a practice task.
Built	Learner submitted an artifact.
Revised	Learner improved after feedback.
Proved	Learner met rubric thresholds and portfolio criteria.

Why this is better than current edtech

Current pattern	OpenEd answer
Video playlists	Interactive lesson workspace.
Static articles	Multimodal learning with tutor and practice.
Generic chatbots	Course-grounded tutor.
Weak certificates	Artifact-based SkillProof.
Educator fame as quality proxy	Course quality and source QA.
Paid AI help	Free learner access plus BYOK.
LMS admin complexity	Learner-first v1 with portals later.

Minimum proof of success

OpenEd should first prove the following with FME:

1. Learners can complete a serious course without feeling lost.
2. The AI tutor helps learners understand, not cheat.
3. Source mapping improves trust.
4. Artifacts make learning feel useful.
5. The educator workflow can create a course with publish-readiness checks.
6. The proof engine can distinguish completion from mastery.

Source notes

This document is grounded in the current OpenEd/FME project context and the following reference set where applicable:

- Uploaded Frontier Model Evaluations prototype code bundle, latest inspected bundle: Design System for Frontier Model Evaluation (5).zip.
- Uploaded Frontier Model Evaluations 51-hour curriculum and Module 1 foundations files.
- Coursera, Udemy, DataCamp, Moodle, Khanmigo, Duolingo public documentation and product pages.
- Supabase Auth and Row Level Security documentation.
- OpenAI API key safety guidance, MDN Web APIs, and YouTube IFrame Player API documentation.

- Bloom's 2 sigma problem, intelligent tutoring systems research, CourseAssist, LearnLM-Tutor, DCCI, and UNESCO guidance for generative AI in education.

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